

Hemaksh Chaturvedi

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EDUCATION

Masters in Computer Science, *New York University*

May 2025 | New York City, USA

Relevant Courses: Machine Learning, Finance Foundation, Data Science, Database Management, Big Data, Algorithm Design

CGPA: 3.71

Bachelor of Technology in Computer Science and Engineering, *Vellore Institute of Technology*

2019 – 2023 | Vellore, India

Relevant Courses: Data Structure and Algorithms, Software Engineering, Data Visualization, Natural Language Processing

CGPA: 8.56

PROFESSIONAL EXPERIENCE

Software Developer / Production Engineering Shift Lead, *Deloitte (via Neotek Solutions Inc)* 📄

Aug 2025 – Present

- Lead production reliability for a distributed ecosystem of **190+ websites and 1,600+ applications**, supporting **8M+ users and 36M monthly requests**, sustaining **90%+ platform availability** through incident response, observability, and platform improvements.
- Designed a health-check-driven **failover architecture** for NLP chatbot services, removing single points of failure and reducing service outages by 30%.
- Rebuilt the authorization layer using **dynamic AWS Lambda configuration**, enabling role-based access without redeployments and reducing operational overhead by **60%**.
- Optimized frontend and backend performance (20+ React components + API integrations), reducing page load latency by **30%** and improving stability.
- Built observability automation (CloudWatch + Lambda) improving incident detection and reducing MTTR by **20%**, saving **25+ hours/month**.
- Serve as escalation lead for production incidents, driving root cause analysis and long-term architectural remediations with **95% on-time resolution**.

Software Developer, *Metropolitan Transportation Authority* 📄

Jan 2025 – Aug 2025

- Led migration of a mission-critical bus yard management platform from Angular to **React + TypeScript**, improving frontend performance by **25%** and reducing technical debt.
- Designed and deployed **stateless RESTful APIs on AWS**, containerized with Docker and automated via CI/CD pipelines, reducing response errors by **18%** and deployment time by **35%**, enabling zero-downtime releases.
- Built an automated API validation pipeline processing **hourly data batches (10K+ records)** using Linux cron and secure S3 storage, eliminating manual checks and reducing validation effort by **80%**, saving **~15 hours/week**.
- Diagnosed and resolved **4 critical API failures** through structured root cause analysis, implementing algorithmic improvements deployed network-wide that improved API data accuracy by **12%** and prevented recurring production incidents.

Quantitative Developer, *Sov AI* 📄

Jun 2024 – Aug 2024

- Built a modular Python library for pairs trading, integrating cointegration, distance-based, and ML-driven selection algorithms; improved strategy returns by **2%** and reduced computation time by **4%** through vectorization and algorithm optimization.
- Designed low-latency signal generation systems using statistical and ML models, improving prediction accuracy by **3%**, reducing execution latency by **8%**.
- Built robust backtesting and risk evaluation frameworks incorporating transaction cost modeling and portfolio optimization.

Machine Learning Fellow, *Fellowship.AI* 📄

Oct 2022 – Jan 2023

- Engineered image processing pipelines using **Python, Trimap, ModNet, and PaddleSeg**, improving illumination segmentation accuracy to **65%**.
- Developed and deployed image harmonization algorithms, enhancing visual consistency and achieving **85% target outcome accuracy**.
- Optimized a diffusion-based deep learning model for scalability, reducing processing time by **20%** and enabling efficient handling of large image datasets.

Full-Stack Developer, *ISPA EXIM PVT LTD* 📄

Aug 2022 – Oct 2022

- Built and optimized a full-stack web platform (HTML/CSS/JS/PHP), increasing traffic by **40%** and reducing page response time by **25%** through backend and query optimizations.
- Designed and executed A/B experiments that improved engagement and conversion by **15%**, driving measurable UX improvements.

PROJECTS

Serverless Photo Search Application (AWS) 📄, *ElasticSearch, Rekognition, S3, API Gateway, CloudFormation*

- Developed a scalable photo album web app utilizing **ElasticSearch, Amazon Lex, and Rekognition**, enabling natural language search. Achieved a **5% reduction in search time** and **improved search accuracy** by allowing users to query photos by people, objects, actions, and landmarks.
- Implemented a **CI/CD pipeline with CloudFormation** for automated deployment and infrastructure provisioning, which resulted in a **8% faster deployment cycle** and **increased deployment reliability by 11%**.

Real-Time Chat Application 📄, *Java, JavaFX, SQLite, Multi-threading, Networking*

- Developed a JavaFX-based real-time chat app with client-server architecture, supporting **20+ concurrent users** with secure authentication and messaging.
- Implemented **multi-threading** to handle **multiple client connections**, reducing latency by **20%** and ensuring smooth communication.
- Integrated **SQLite** for persistent data storage, enabling retrieval of **1000+ chat messages** and ensuring reliable data management.

SKILLS

Languages: Python, Java, C++, C#, C, JavaScript, TypeScript, React, R, ReactJS

Backend & APIs: RESTful Services, Spring Boot, Flask, Microservices Architecture, Multi-threading, GraphQL

Cloud & DevOps: AWS (Lambda, S3, CloudWatch, WAF, ElasticSearch, Lex, Rekognition, API Gateway, CloudFormation), Docker, CI/CD Pipelines

Databases: PostgreSQL, MySQL, MongoDB, SQL, Distributed systems fundamentals, Performance optimization, schema design

AI/ML: TensorFlow, PyTorch, NLP systems, Model Optimization

Tools: Git, Github, Jira, Confluence, Agile, Technical documentation

ACHIEVEMENTS

IMC Trading - Prosperity 2 📄, *Manual and Algorithmic Trading Competition*

Apr 2024

- Achieved 93rd rank in India, 239th rank in the USA and overall rank of 873rd globally